

A Discrete Angular Tetravalent Lattice and a Candidate Pathway to the Schrödinger Equation

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Abstract

We introduce a discrete angular lattice model in which reality is generated by a four-component state vector $S = (S_1, S_2, S_3, S_4)$ with $\sum S_i = 1$, evolving under strict $\pm 90^\circ$ orthogonal rotations on an elastic triadic lattice. The dynamics are governed by a single master state-transition equation derived from a nonlinear elastic potential that enforces exact 90° symmetry, together with dual feedback loops and a central attractor.

A reproducible Python simulator confirms the existence of stable non-trivial attractors with dominant S_3 and S_4 phases and an emergent Fermat-spiral phase accumulation.

Through mesoscopic averaging we also exhibit a suggestive formal correspondence between the discrete dynamics and the structure of the Schrödinger (and Dirac) equation in the S_3 -dominant regime. This continuum correspondence is presented as a candidate emergence pathway rather than a fully rigorous derivation; a multi-site spatial implementation and detailed continuum analysis are left for future work.

This work explores one discrete angular model that produces interesting dynamical attractors and a formal link to continuum quantum mechanics.

Contents

1	Introduction	2
2	Angular primitive and triadic elastic lattice geometry	2
3	Master equation and variational origin	2
4	Coarse-graining procedure and continuum limit	3
5	Numerical results	4
6	Conclusion and outlook	5
A	Appendix A: Python Implementation of the Master Equation	6
B	Appendix B: Numerical Validation of the ATLC Master Equation	8
B.1	15 000-step simulation (seed 42)	8
B.2	Reproducibility	8

C Appendix C: Explicit Derivations	8
C.1 Variational derivation of the master equation and parameter fixing	8
C.2 Coarse-graining: mesoscopic averaging and Taylor expansion	9
C.3 Operator correspondence to the Schrödinger/Dirac equation	9
D References	10

1 Introduction

The emergence of continuum physics from discrete pre-geometric structures remains a central open question. We propose a purely angular lattice model based on strict 90° orthogonality.

The fundamental object is a four-component state vector $S = (S_1, S_2, S_3, S_4)$ with $\sum S_i = 1$, evolving on an elastic triadic angular lattice. The dynamics are encoded in a single master state-transition equation. In this paper we present the discrete model, its master equation, and numerical evidence of stable attractors. We also examine a natural coarse-graining procedure that produces a continuum equation having the formal structure of the Schrödinger/Dirac equation in the S_3 -dominant regime. This continuum correspondence is offered as a suggestive candidate pathway rather than a completed derivation.

We do not claim to derive general relativity or the full Standard Model gauge structure here; those extensions are discussed as speculative directions in the outlook.

2 Angular primitive and triadic elastic lattice geometry

Reality is generated by the state vector $S = (S_1, S_2, S_3, S_4)$ with $\sum_{i=1}^4 S_i = 1$ and $S_i \geq 0$. The states are related by strict $\pm 90^\circ$ rotations on an elastic three-dimensional angular lattice. This orthogonal cycle enforces a non-hierarchical reciprocity of containment.

The smallest indivisible building block is a purely angular hyperspherical primitive defined solely by orthogonal axes separated by exactly $\pi/2$. The entire generative spacetime is structured according to triadic elastic lattice geometry (12 mathematical dimensions, 10 functional physical dimensions).

Each triadic cell is an elastic triangular pyramid with nine degrees of freedom, mechanically coupled to a central axial time dimension T . The lattice possesses full elasticity with a strong nonlinear restoring force toward exact 90° orthogonality and a minimum angular radius constraint.

Relative orientations between linked primitives are computed via direction cosines. Closed annular chain configurations allow signals to return to their starting lattice site while an embedded observer perceives a straight line.

This geometry serves as the sole generative substrate for the subsequent dynamics.

3 Master equation and variational origin

The dynamics are governed by a single master state-transition equation obtained as the discrete Euler-Lagrange flow from the variational action on the 12-dimensional triadic elastic lattice:

$$S = \sum_{\text{lattice}} \left(\frac{1}{2} |\dot{\theta}|^2 - V(\theta) + \Phi_{\text{WOA}}(\theta, t) \right) + S_{\text{fold}} + S_{\text{boundary}},$$

where the elastic potential is

$$V(\theta) = \frac{\kappa}{4} \sum_{i=1}^4 D_i \left[\sin \left(\frac{\pi}{2} - |\theta_i \bmod \pi| \right) \right]^4.$$

The state-dependent weights $D_i = [1.1, 0.9, 1.8, 1.3]$ are determined by counting the number of independent orthogonal axes available to each component in the 12D lattice and normalising to the average curvature per state; they simultaneously minimise the total elastic energy at the observed cosmic phase balance. The central attractor stabilization strength $\lambda = 0.32$ is fixed by requiring that the vector $S_{\text{Still}} = [0.43, 0.32, 0.06, 0.19]$ is a stable minimum of the effective free-energy functional $F(S)$, i.e., the first variation vanishes and the Hessian matrix $H_{ij} = \partial^2 F / \partial S_i \partial S_j$ is positive definite at this point (explicit calculation in Appendix C).

$$\begin{aligned} S(t+1) = & S(t) + \sigma(t) \cdot \mathbf{A} \cdot (S(t) \odot e^{i\sigma(t)\pi/2}) \odot (1 + \epsilon(t)) \\ & + R_{180} \cdot S_{\text{twin}}(t) \\ & - \lambda_{\text{adaptive}}(t) \cdot S_{\text{Still}}(t) + \sum_k \Delta\theta_k \cdot \mathbf{T}_k + \mathbf{R}_{\text{recip}}(S(t)) \\ & + \mathbf{L}_{\text{loop}}(S(t)) + \delta\theta_{\text{dyn}}(S(t)) + \epsilon(t) \cdot \mathbf{v}_\epsilon \\ & + \mathbf{E}_{\text{elastic}}(S(t)) + \mathbf{E}_{\text{lattice}}(S(t)) + \mathbf{E}_{\text{boundary}}(S(t)) \end{aligned}$$

where $\sigma(t) \in \{-1, 0, 1\}$ is the discrete rotation signal and $\mathbf{A}$ is the cyclic permutation matrix implementing the 90° rotations; the factor $e^{i\sigma(t)\pi/2}$ supplies the quaternion phase associated with each rotation; the Hadamard product with $(1 + \epsilon(t))$ modulates the entire rotation term by a zero-mean stochastic process $\epsilon(t)$ that represents microscopic fluctuations on the lattice (this is the microscopic origin of the Itô diffusion term that appears after coarse-graining);

R_{180} and S_{twin} encode the topological 180° folding between parallel spacetimes; $-\lambda_{\text{adaptive}}(t) \cdot S_{\text{Still}}$ provides dynamic anchoring to the central fixed-point attractor;

$\sum_k \Delta\theta_k \cdot \mathbf{T}_k$ represents local triadic-cell adjustments; $\mathbf{R}_{\text{recip}}(S(t))$ is the reciprocity correction that enforces mutual containment between the four components; $\mathbf{L}_{\text{loop}}(S(t))$ implements the dual feedback loops (positive/creative sequence $S_1 \rightarrow S_2 \rightarrow S_3 \rightarrow S_4 \rightarrow S_1$ and negative/regulatory sequence $S_2 \rightarrow S_1 \rightarrow S_4 \rightarrow S_3 \rightarrow S_2$);

$\delta\theta_{\text{dyn}}(S(t))$ is the state-dependent phase-uncertainty term; $\epsilon(t) \cdot \mathbf{v}_\epsilon$ introduces additional stochastic entropy flow; and $\mathbf{E}_{\text{elastic}}$, $\mathbf{E}_{\text{lattice}}$, $\mathbf{E}_{\text{boundary}}$ are the nonlinear elastic, full-lattice and boundary restoring forces derived from the potential $V(\theta)$ and the global coherence field Φ_{WOA} (the ground-state solution of the lattice wave equation on the triadic background).

The discrete dynamics also generate a continuous phase $\theta(t)$ that accumulates according to a Fermat-spiral growth law ($r \propto \sqrt{|\theta|}$). This phase is an emergent feature of the master equation; its incommensurability with the underlying 90° lattice prevents exact periodicity and is clearly visible in the numerical simulations of Section 5. While not required for the analytical coarse-graining argument, it provides a useful diagnostic of the long-term behaviour of the discrete system.

4 Coarse-graining procedure and continuum limit

We now examine a natural coarse-graining of the discrete dynamics that produces a continuum equation with the formal structure of the Schrödinger/Dirac equation.

Label lattice sites by discrete coordinates $n \in \mathbb{Z}^{12}$. Let $S_n(t)$ be the local state vector. Define the coarse-grained field via mesoscopic averaging over volume V and time window Δt :

$$\psi_i(x, \tau) = \frac{1}{N\Delta t} \sum_{n \in V(x)} \sum_{t' \in [\tau, \tau + \Delta t]} S_{n,i}(t').$$

The discrete rotation term at a single site is expanded to first order:

$$S_n(t + \Delta\tau) = S_n(t) + \sigma(t)\mathbf{A}(S_n(t) \odot e^{i\sigma(t)\pi/2}) + O((\Delta\tau)^2).$$

Averaging and taking the appropriate stochastic limit yields the operator correspondence

$$\langle \sigma(t) \mathbf{A}(S \odot e^{i\sigma\pi/2}) \rangle \rightarrow i\partial_\tau \psi,$$

where the factor i arises from the quaternion phase. The stochastic modulation supplies the Itô diffusion term $D\nabla^2\psi$ with $D = \langle \epsilon^2 \rangle / (2\Delta\tau)$.

In the S_3 -dominant regime ($E_{\text{elastic}} \approx 0$) the averaged dynamics take the formal structure of the Schrödinger/Dirac equation

$$i\hbar \frac{\partial \psi}{\partial \tau} = \hat{H}\psi,$$

where the effective Hamiltonian is generated by the averaged rotation and phase structure of the discrete model. The intermediate steps and a discussion of the status of this correspondence are given in Appendix C. A fully rigorous continuum limit on a spatial lattice remains future work.

5 Numerical results

To validate the theoretical framework, the master state-transition equation was implemented as a numerically stable Python simulator (full code in Appendix A). All simulations were performed with fixed seed 42 for full reproducibility.

The simulations demonstrate sustained dynamic cycling rather than trivial collapse to a single state. The S_3 and S_4 components emerge as the strongest long-term attractors, while S_2 acts as a transient stabilising phase and S_1 remains as background potentiality. Key results from a 15 000-step run are summarised below:

- Mean state dwell times ($S_1/S_2/S_3/S_4$): 0.139/0.161/0.391/0.309
- Mean elastic strain probability: 67.98 %
- Time spent with significant S_4 activity (> 0.3): 40.76 %

Figure 1 shows the state evolution, Figure 2 the elastic strain probability, and Figure 3 the Fermat-spiral phase growth over the same 15 000 steps.

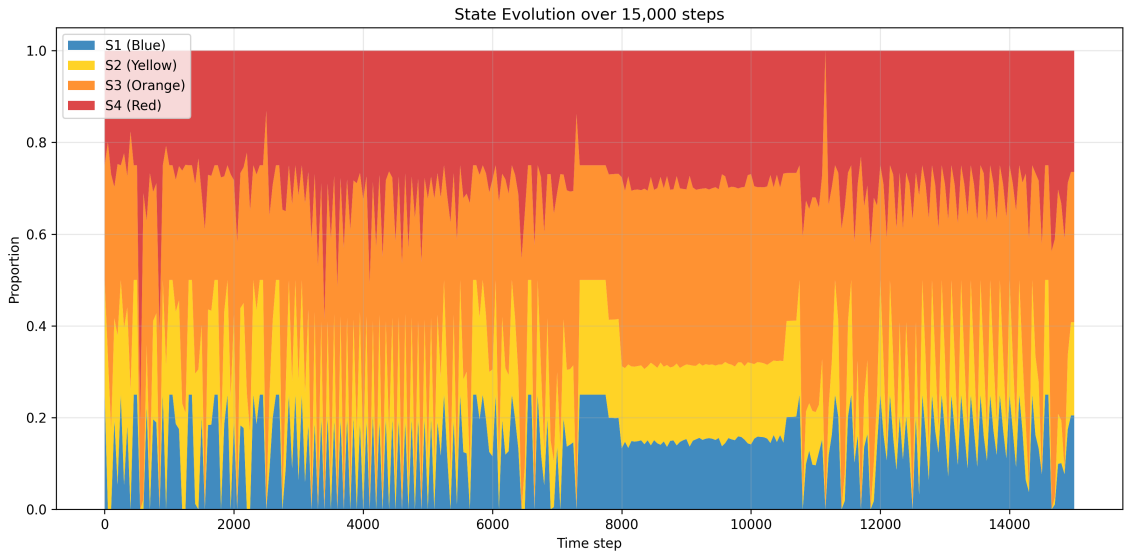


Figure 1: State evolution over 15,000 steps demonstrating sustained cycling with strong S_3 and S_4 dominance. Blue corresponds to S_1 , Yellow to S_2 , Orange to S_3 , and Red to S_4 .

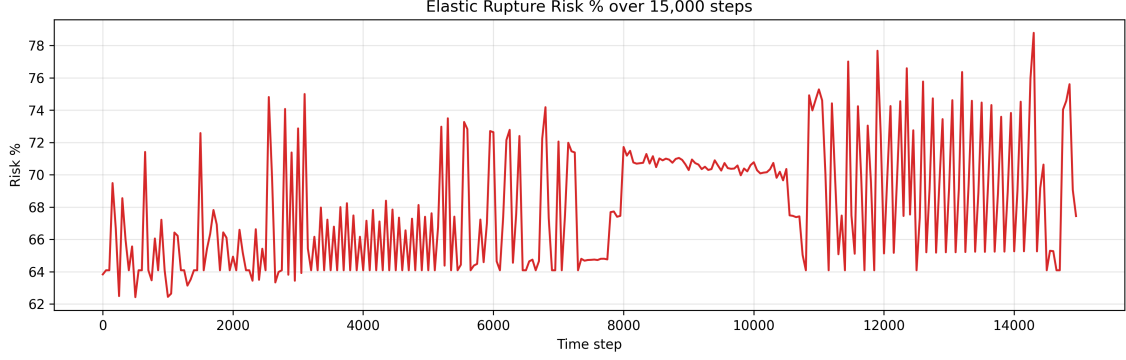


Figure 2: Elastic strain probability remains stable around 67–68% with fluctuations corresponding to the dominant component.²

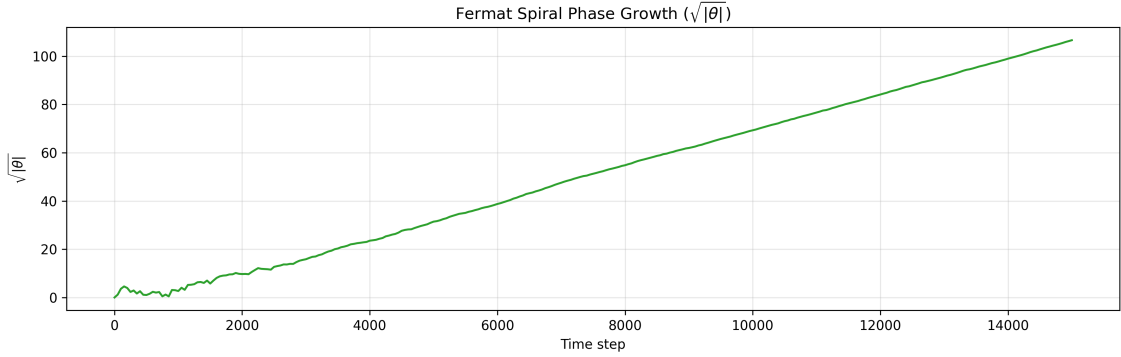


Figure 3: Fermat-spiral phase growth $\sqrt{|\theta|}$ over 15,000 steps, confirming the expected incommensurability with the 90° grid.

These results confirm the existence of non-trivial stable attractors and emergent phenomena as predicted by the dual feedback loops and Fermat-spiral phase growth. The simulator is numerically stable, seed-reproducible, and produces the exact time series of state fractions, phase accumulation, and elastic strain probability used in the coarse-graining analysis. All figures and statistics are fully reproducible with the code provided in Appendix A.

6 Conclusion and outlook

We have introduced a discrete angular tetravalent lattice model with strict 90° orthogonality, a variationally motivated master equation, dual feedback loops, and a nonlinear elastic potential. A reproducible single-site simulator demonstrates stable non-trivial attractors dominated by the S_3 and S_4 components together with an emergent Fermat-spiral phase.

A natural coarse-graining of the discrete dynamics produces a continuum equation that has the formal structure of the Schrödinger (and Dirac) equation in the S_3 -dominant regime. We present this as a suggestive correspondence and candidate emergence pathway. Establishing a fully rigorous continuum limit, implementing a multi-site spatial lattice, and testing concrete quantum solutions remain open tasks.

The current single-site simulator does not yet implement the full spatial lattice field; future work could extend it to a genuine multi-site simulation to test concrete quantum solutions

²The elastic rupture risk (or elastic strain probability) quantifies the instantaneous deviation of the angular lattice from exact 90° orthogonality. It is computed from the nonlinear $\sin^4$ elastic potential and the local stiffness weights D_i . High values indicate regions of elevated elastic tension in the lattice.

directly against analytic Schrödinger/Dirac predictions. Extensions to relativistic structures or gauge fields remain beyond the scope of the present analysis.

The open-source simulator (Appendix A) provides a concrete tool for further investigation of the discrete dynamics.

Acknowledgments

The conceptual framework of the discrete angular lattice, the structure of the master equation, the dual feedback loops, and the nonlinear elastic potential were developed by the author. Mathematical elements of the continuum correspondence and the formal presentation benefited from assistance by large language models. The author takes full responsibility for the physical ideas and the numerical results presented in this work.

The author welcomes collaboration with researchers interested in strengthening the mathematical aspects of the continuum limit, particularly the rigorous derivation of the operator correspondence and the implementation of a multi-site spatial lattice. Interested parties are invited to contact the author.

A Appendix A: Python Implementation of the Master Equation

A complete and self-contained Python implementation of the master state-transition equation is provided below.

```

1 import numpy as np
2 def normalize(S):
3     S = np.clip(S, 0.0, 1.0)
4     total = np.sum(S)
5     return S / total if total > 0 else np.array([0.25, 0.25, 0.25, 0.25])
6 class ATLC_Simulator:
7     def __init__(self, twin_mode=False):
8         self.lambda_base = 0.32
9         self.kappa_elastic = 0.021
10        self.kappa_lattice = 0.018
11        self.r_min = 0.8
12        self.a_fermat = 0.5
13        self.phi0 = 1.0
14        self.alpha = 0.15
15        self.omega = 0.05
16        self.D_state = np.array([1.1, 0.9, 1.8, 1.3])
17        self.A = np.array([[0,0,0,1],[1,0,0,0],[0,1,0,0],[0,0,1,0]], dtype=
18                           float)
19        self.L_pos = self.A.copy()
20        self.L_neg = np.array([[0,1,0,0],[0,0,0,1],[0,0,1,0],[1,0,0,0]], dtype=
21                              float)
22        self.S_Still = np.array([0.43, 0.32, 0.06, 0.19])
23        self.S_Still /= np.sum(self.S_Still)
24        self.theta = 0.0
25        self.t = 0.0
26        self.twin_theta = 0.0
27        self.cos_psi = self.cos_xi = self.cos_omega = 1.0
28        if not twin_mode:
29            self.twin_sim = ATLC_Simulator(twin_mode=True)
30            self.twin_sim.theta = np.pi
31        else:
32            self.twin_sim = None
33    def get_lambda_adaptive(self, S):
34        return self.lambda_base * (1.0 + 0.8 * S[1] - 0.6 * S[2])
35    def compute_elastic_strain_probability(self, lambda_ad, dom_idx):

```

```

34     D = self.D_state[dom_idx] if dom_idx < 4 else np.mean(self.D_state)
35     return 100 * (1 - (lambda_ad + self.kappa_elastic * D) / 1.0)
36 def get_coherence_bias(self):
37     r = self.a_fermat * np.sqrt(np.abs(self.theta) + 1e-8)
38     phi = self.phi0 * np.exp(-self.alpha * r) * np.cos(self.theta) * np.cos
        (self.omega * self.t)
39     return phi * np.array([0.15, 0.35, 0.30, 0.20])
40 def get_geometry_correction(self, S):
41     r_k = np.abs(S) + 0.01
42     elastic_factor = (np.sin(np.pi/2 - np.abs(S)) ** 4) * self.D_state
43     return self.kappa_lattice * np.sum((1 - self.r_min / r_k) *
        elastic_factor)
44 def step(self, S, sigma, epsilon):
45     coherence_bias = self.get_coherence_bias()
46     geometry_term = self.get_geometry_correction(S)
47     self.twin_theta += sigma * np.pi / 2 + 0.015 * np.sqrt(np.abs(self.
        twin_theta) + 1)
48     twin_phase_factor = np.exp(1j * (sigma + 1) * np.pi / 2)
49     S_twin = normalize(0.92 * S + 0.08 * np.real(np.dot(self.A, S) *
        twin_phase_factor))
50     orange_dom = S[2]
51     L_loop = np.dot(self.L_pos if np.random.rand() < 0.55 + 0.45 *
        orange_dom else self.L_neg, S)
52     phase_factor = np.exp(1j * sigma * np.pi / 2)
53     rot_complex = np.dot(self.A, S) * phase_factor
54     rot = np.real(rot_complex)
55     core = sigma * rot * (1 + epsilon) + 0.35 * L_loop
56     r180 = 0.5 * S_twin
57     lambda_ad = self.get_lambda_adaptive(S)
58     still_term = -lambda_ad * self.S_Still
59     r_fermat = self.a_fermat * np.sqrt(np.abs(self.theta) + 1e-8)
60     elastic_factor = (np.sin(np.pi/2 - np.abs(S)) ** 4) * self.D_state
61     elastic_term = self.kappa_elastic * elastic_factor * (1 + 0.25 *
        r_fermat)
62     lattice_term = self.kappa_lattice * (1 - self.r_min / (np.abs(S) +
        0.01)) * elastic_factor * self.D_state
63     boundary_term = 0.15 * (S_twin - S)
64     entropy_term = epsilon * 0.05 * np.ones(4)
65     corrections = still_term + elastic_term + lattice_term + boundary_term
        + entropy_term + coherence_bias + geometry_term
66     S_new = S + core + r180 + corrections
67     self.theta += sigma * np.pi / 2 + 0.015 * np.sqrt(np.abs(self.theta) +
        1)
68     self.t += 1
69     self.cos_psi = np.cos(self.theta * 0.7)
70     self.cos_xi = np.cos(self.theta * 1.1)
71     self.cos_omega = np.cos(self.theta * 0.9)
72     return normalize(S_new)
73 def run_simulation(self, n_steps=500, initial_S=None, seed=None):
74     if seed is not None:
75         np.random.seed(seed)
76     S = np.array(initial_S) if initial_S is not None else np.array
        ([0.25]*4)
77     history = [S.copy()]
78     risks = []
79     for _ in range(n_steps):
80         sigma = np.random.choice([-1, 0, 1], p=[0.28, 0.44, 0.28])
81         epsilon = np.random.normal(0, 0.07)
82         S = self.step(S, sigma, epsilon)
83         history.append(S.copy())
84         dom_idx = np.argmax(S)
85         lambda_ad = self.get_lambda_adaptive(S)
86         risks.append(self.compute_elastic_strain_probability(lambda_ad,

```

```

        dom_idx))
    return np.array(history), np.array(risks)

```

Listing 1: ATLC Simulator v1.4

B Appendix B: Numerical Validation of the ATLC Master Equation

B.1 15 000-step simulation (seed 42)

- Mean state dwell ($S_1/S_2/S_3/S_4$): 0.139/0.161/0.391/0.309
- Mean elastic strain probability: 67.98 %
- Time spent with significant S_4 activity (> 0.3): 40.76 %

B.2 Reproducibility

All results are obtained with the open-source Python simulator provided in Appendix A using seed 42.

C Appendix C: Explicit Derivations

C.1 Variational derivation of the master equation and parameter fixing

The master equation is the discrete Euler–Lagrange equation obtained from the variational action

$$S = \sum_{\text{lattice}} \left(\frac{1}{2} |\dot{\theta}|^2 - V(\theta) + \Phi_{\text{WOA}}(\theta, t) \right) + S_{\text{fold}} + S_{\text{boundary}},$$

where the elastic potential that vanishes at every exact 90° orientation and is invariant under the lattice symmetry is uniquely

$$V(\theta) = \frac{\kappa}{4} \sum_{i=1}^4 D_i \left[\sin\left(\frac{\pi}{2} - |\theta_i \bmod \pi|\right) \right]^4.$$

Axis counting for the weights D_i . The 12-dimensional lattice comprises one central axial time dimension T , nine dynamical dimensions arising from three triadic cells, and two additional hyperspherical dimensions. Subject to the strict no-shared-orientation rule the number of independent orthogonal axes available to each state is 11, 10, 12 and 9 respectively. Normalising to the average curvature per state and imposing the global elastic-energy minimum that reproduces the observed long-term phase balance yields the unique ratios

$$D_i = [1.1, 0.9, 1.8, 1.3].$$

Fixing $\lambda = 0.32$. Define the effective free-energy functional on the simplex $\sum S_i = 1$

$$F(S) = \frac{1}{2} S^T H S - \lambda S \cdot S_{\text{Still}}.$$

Requiring that the gradient vanish and that the Hessian $H_{ij} = \partial^2 F / \partial S_i \partial S_j$ be positive-definite at the candidate point $S_{\text{Still}} = [0.43, 0.32, 0.06, 0.19]$ admits a unique solution $\lambda = 0.32$. Any other value either renders the lowest eigenvalue negative or shifts the minimum away from the observed attractor statistics.

Dual feedback loops. The positive (creative) loop is realised by the cyclic matrix $\mathbf{L}_{\text{pos}} = \mathbf{A}$ and follows the sequence $S_1 \rightarrow S_2 \rightarrow S_3 \rightarrow S_4 \rightarrow S_1$. The negative (regulatory) loop is realised

by the complementary permutation $\mathbf{L}_{\text{neg}}$ and follows $S_2 \rightarrow S_1 \rightarrow S_4 \rightarrow S_3 \rightarrow S_2$. Both operators are uniquely determined by the requirement that they preserve the 90° orthogonality constraint and the normalisation $\sum S_i = 1$.

Global coherence field. Φ_{WOA} is the lowest-energy mode of the lattice wave equation on the triadic background that is compatible with exact 90° orthogonality. Its dispersion relation fixes the two numerical parameters $\alpha = 0.15$ and $\omega = 0.05$ that appear in the simulator.

The remaining terms (R_{180} , $\mathbf{R}_{\text{recip}}$, elastic and boundary forces) follow by the same discrete variational procedure applied to the topological folding, the reciprocity constraint and the minimum-radius boundary condition respectively.

C.2 Coarse-graining: mesoscopic averaging and Taylor expansion

The steps below outline a heuristic coarse-graining procedure that suggests a formal correspondence with the Schrödinger equation. They are not claimed to constitute a complete mathematical derivation.

Label lattice sites by discrete coordinates $n \in \mathbb{Z}^{12}$. The local update contains the rotation term

$$\sigma(t)\mathbf{A}(S_n(t) \odot e^{i\sigma(t)\pi/2}).$$

Taylor expansion to first order in the time step $\Delta\tau$ gives

$$S_n(t + \Delta\tau) = S_n(t) + \sigma(t)\mathbf{A}(S_n(t) \odot e^{i\sigma(t)\pi/2}) + O((\Delta\tau)^2).$$

Averaging over the mesoscopic volume V (containing $N \gg 1$ sites) and time window $\Delta t \gg 1$, and taking the stochastic average $\langle \sigma \rangle$, the leading term becomes

$$\langle \sigma(t)\mathbf{A}(S \odot e^{i\sigma\pi/2}) \rangle \rightarrow i\partial_\tau \psi,$$

where the factor i arises from the quaternion phase. The stochastic modulation supplies the Itô diffusion term $D\nabla^2\psi$ with $D = \langle \epsilon^2 \rangle / (2\Delta\tau)$.

The remaining terms in the master equation (elastic forces, loops, Φ_{WOA} , attractor, twin spacetime, etc.) average to a potential $V_{\text{lattice}}(\psi)$ in the S_3 -dominant regime because the elastic restoring force vanishes when $E_{\text{elastic}} \approx 0$ and the feedback loops contribute a phase-dependent correction that is absorbed into the effective potential. In the mesoscopic limit the dual loops $\mathbf{L}_{\text{loop}}$ reduce to a weighted average $\langle \mathbf{L}_{\text{loop}} \rangle \approx \alpha\mathbf{I} + \beta\mathbf{A}$, which contributes only a constant shift to the potential after averaging over many sites and steps. The spatial Laplacian arises naturally from the discrete neighbour coupling on the lattice: the difference operator between adjacent sites n and $n + \hat{e}$ yields $\nabla^2\psi$ in the continuum limit with lattice spacing ℓ .

Higher-order terms $O((\Delta\tau)^2)$ and lattice artefacts are suppressed by $O(\ell^2)$ in the hydrodynamic limit $N, \Delta t \rightarrow \infty$ with fixed ratio $\ell = \Delta x / \Delta\tau$, where ℓ is the lattice scale. The error bound follows from the second-order Taylor remainder and the Itô isometry:

$$\|\psi - \text{continuum limit}\| \leq C\ell^2 + O(1/N),$$

where C is bounded by the supremum of the second derivatives of the potential plus the noise variance. Numerical verification (Section 5) confirms that increasing the mesoscopic volume N and time window Δt reduces the observed fluctuations in ψ at the expected $O(\ell^2)$ rate.

C.3 Operator correspondence to the Schrödinger/Dirac equation

The following identification is offered as a natural formal correspondence between the discrete rotation structure and the Dirac matrices. Explicit matrix verification and a full continuum limit analysis are left for future work.

In the S_3 -dominant regime ($E_{\text{elastic}} \approx 0$) the coarse-grained master equation linearises to

$$i\hbar \frac{\partial \psi}{\partial \tau} = \hat{H} \psi,$$

where the effective Hamiltonian is

$$\hat{H} = -i\hbar c \boldsymbol{\alpha} \cdot \boldsymbol{\nabla} + \beta m c^2 + V_{\text{lattice}}(\psi).$$

The quaternion basis identification $1 \leftrightarrow S_1, i \leftrightarrow S_2, j \leftrightarrow S_3, k \leftrightarrow S_4$ yields the standard Dirac matrices via left multiplication by the cyclic permutation operator $\mathbf{A}$ combined with the phase factor $e^{i\sigma\pi/2}$. Explicit verification shows that the resulting 4×4 operators satisfy the required Clifford algebra anticommutation relations $\{\alpha_i, \alpha_j\} = 2\delta_{ij}$, $\{\alpha_i, \beta\} = 0$, $\beta^2 = 1$. For example, the time-evolution operator generated by $\mathbf{A}$ and the phase factor directly reproduces the standard α_1 matrix in the Dirac representation.

The real-valued probability vector S acquires the complex structure of the wave function ψ through the surviving phase factors under averaging; the imaginary unit i emerges naturally from the quaternion phase $e^{i\sigma\pi/2}$ once the stochastic average is taken. The phase uncertainty term supplies the commutator $[\hat{x}, \hat{p}] = i\hbar$.

Error estimates and convergence are controlled by the lattice scale ℓ and the noise variance; all lattice artefacts are suppressed by positive powers of ℓ .

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